

XRPL OPTIONS SIDECHAIN

The Native Derivatives Layer for the XRP Ledger

“Hyperliquid proved that a purpose-built chain with a native order book can dominate DeFi derivatives. XRPL is next.”

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github.com/Transia-RnD/rippled • branch: options-sidechain

The Opportunity

Crypto derivatives are a multi-trillion dollar market. On-chain, it is almost entirely dominated by centralized exchanges and a handful of chains that weren't built with derivatives in mind.

Hyperliquid changed the calculus. By building a purpose-built L1 with a native order book baked into the chain itself, they captured billions in daily volume, a \$9B+ token valuation, and the mindshare of the most serious traders in DeFi — without VC money, without compromises.

XRPL has never had an equivalent. The XRP Ledger is one of the fastest, lowest-fee, most battle-tested blockchains in existence — with a native DEX, a global validator network, and nearly 15 years of uptime. It has processed billions of transactions. But it has no derivatives layer.

That gap ends here.

The XRPL Options Sidechain is a purpose-built trading chain that brings American-style options, up to 200x leveraged margin trading, and a trustless cross-chain bridge to the XRP Ledger ecosystem — secured by the same validator network that secures XRPL today.

Why This, Why Now

The Derivatives Gap

Options are the most powerful financial instrument in existence. In traditional finance, the global options market dwarfs futures. In crypto, options volume is dominated by Deribit — a centralized exchange. On-chain options are nascent, technically hard, and in high demand from institutional and professional traders.

Hyperliquid proved perpetual futures can be done on-chain at CEX quality. No one has done the same for options. Yet.

The XRPL Advantage

XRPL is not a general-purpose chain trying to do everything. Like Hyperliquid, it was built with specific performance characteristics in mind: sub-4-second finality, near-zero fees, a native order book, and a federated validator model that enables exactly the kind of trusted-committee design this sidechain requires.

The XRPL community has deep liquidity in XRP and a growing ecosystem of tokenized assets via MPTokens. A derivatives layer captures that value and puts it to work.

Institutional Readiness

This is not a toy. The Options Sidechain includes native WebAuthn/FIDO2 passkey support — meaning traders can sign transactions with Face ID, Touch ID, or hardware security keys. This is the authentication standard used by banks and enterprises worldwide. It removes one of the biggest UX and security barriers to institutional on-chain trading.

How It Works

Three interlocking systems, all implemented natively at the protocol level:

1. Trustless Cross-Chain Bridge (XPop Import/Export)

Assets move between XRPL mainnet and the sidechain via XPop — a cryptographic proof that a transaction was validated and included in a specific mainnet ledger, signed by 80%+ of the validator set.

Step	Action	Chain
1	User sends XRP to vault (multisig)	XRPL Mainnet
2	XPop proof generated from ledger	Mainnet Ledger
3	User submits Import with XPop blob	Sidechain
4	Sidechain mints XRP 1:1	Sidechain
5	User submits Export to burn XRP	Sidechain
6	Witnesses release XRP from vault	XRPL Mainnet

Key properties:

- 1:1 lock-and-mint model — total sidechain supply never exceeds vault balance
- No XRPL mainnet amendments required — vault is a standard multisig account
- 80%+ validator quorum required for every import proof
- Full replay protection at account and validator list level
- New accounts bootstrapped automatically on first import

The bridge requires no changes to XRPL mainnet. Validators run the sidechain and serve as the trusted witness set. This is exactly how Hyperliquid's ETH bridge works — two-thirds of staking power must approve withdrawals.

2. Native Options and Margin Trading

American-style option contracts, fully on-chain. Not AMM-based. Not synthetic. Real matched counterparty positions settled at the protocol level.

Feature	Details
Option types	Calls and Puts, American-style
Settlement	Cash settlement on exercise
Naked options	Uncollateralized sell-side (unlimited risk/reward)
Covered options	Collateral locked at contract creation
Leverage	2x to 200x via margin system

Margin modes	Isolated (per-position) or Cross-margin (shared pool)
Liquidation	Open to any account — incentivized with bonus
Insurance	LP-backed insurance vaults per market
Funding rates	Periodic long/short rebalancing mechanism
Order matching	Central Limit Order Book, natively on-chain

The Margin Mathematics

The system uses rigorous financial math, identical to what institutional perp exchanges use:

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Initial Margin = Notional Value / Leverage
Liquidation Price (Long) = Entry × (1 - 1/Leverage + Maintenance Rate)
Unrealized PnL (Long) = (Mark - Entry) × Size
Health = Position Equity ≥ Maintenance Margin
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Why Options Beat Perps for Sophisticated Traders

Capability	Perps (Hyperliquid)
Direction trading	✓
Defined risk for buyers	✗
Volatility trading	✗
Complex multi-leg strategies	✗
Institutional demand	Moderate
Market maturity in crypto	Saturated

Capability	Perps (e.g. Hyperliquid)	Options (This System)
Direction trading	✓	✓
Defined-risk buying	✗	✓
Volatility trading	✗	✓
Complex strategies	✗	✓
Institutional demand	Moderate	Very High

On-chain competition	Saturated	Nascent
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3. Passkey Authentication (WebAuthn/FIDO2)

The sidechain introduces P256 (secp256r1) as a native key type, enabling WebAuthn/FIDO2 passkey authentication for XRPL accounts. Users can sign transactions with:

- Face ID or Touch ID on iPhone / Mac / Android
- Windows Hello
- YubiKey or other FIDO2 hardware security keys

This is the authentication standard used by major banks, Google, and Apple. It replaces seed phrases with device-native biometrics — without any custodial risk. Accounts can register up to 10 passkey credentials, enabling hardware key backup and enterprise multi-device setups.

For institutional traders, this is not a feature — it is a requirement. Passkey support removes the single biggest operational security risk in on-chain trading: private key management.

The Validator Ask

This sidechain needs trusted validators. We are looking for established XRPL UNL validators willing to extend their infrastructure to run a sidechain node.

What we're asking:

Run a UNL validator node for the XRPL Options Sidechain.

Your validator key becomes a signer in the bridge's trust model. When users import XRP from mainnet, your validator's signature is part of the 80%+ quorum that proves the transfer is legitimate.

What Validators Do

- Run a sidechain rippled node (same software stack you already know)
- Participate in consensus for the options sidechain
- Serve as witnesses: observe ExportRecord events and co-sign vault releases on mainnet
- Maintain 80%+ uptime to ensure bridge liveness

Why Be an Early Validator

Hyperliquid launched with 16 validators. It now has a \$9B+ ecosystem. The validators who were there at the start were not just infrastructure providers — they were founding stakeholders.

XRPL has no on-chain derivatives infrastructure today. The validators who help launch this sidechain will be the foundation of the first native options market on one of the oldest and most liquid blockchains in crypto.

Early validators are recognized in the genesis configuration, establishing provenance and trust within the ecosystem from day one.

Technical Requirements

Requirement	Detail
Software	rippled (options-sidechain branch)
Hardware	Standard XRPL validator spec (4+ vCPU, 16GB RAM, SSD)
Network	Stable internet, static IP
Existing UNL	Preferred (establishes validator reputation)
Commitment	Sidechain UNL participation + witness duties
Coordination	Join validator coordination channel for genesis config

XRPL Options vs. The Landscape

	Deribit	Hyperliquid	Lyra / Hegic	XRPL Options
On-chain	✗	✓	✓	✓
Options (not just perps)	✓	✗	✓	✓
Native order book	✗	✓	✗	✓
Leverage (margin)	✓	✓	Limited	✓
Naked options	✓	N/A	✗	✓

Passkey / WebAuthn	✗	✗	✗	✓
Trustless bridge	✗	✓	Varies	✓
XRPL ecosystem	✗	✗	✗	✓

Specification Summary

Three XLS draft specifications define this system:

Spec	Description
XLS-XX: Import/Export	XPop bridge — lock-and-mint / burn-and-release with 80%+ validator quorum
XLS-62: Options & Margin	American options, 2-200x leverage, isolated/cross margin, insurance vaults, liquidation, funding rates
XLS-XX: Passkey Auth	P256/WebAuthn signing — Face ID, Touch ID, hardware keys for transaction signing

Full specifications are available at:

github.com/Transia-RnD/rippled/tree/options-sidechain/specs

Security Audit

This is protocol-level financial infrastructure. Before mainnet launch, every component will undergo independent professional security review.

We are actively seeking a professional security audit firm with experience in both blockchain protocol security and financial systems. The audit will be grant-funded through the XRPL Grants program and scoped across three systems:

System	Audit Focus Areas	Risk Level
XPop Import/Export Bridge	Replay attacks, validator collusion, quorum manipulation, vault security, XPop proof forgery, cross-chain double-spend	Critical

Options & Margin Engine	Liquidation math correctness, margin invariant enforcement, oracle manipulation, insurance vault solvency, naked option settlement edge cases	Critical
Passkey / P256 Auth	P256 implementation correctness, WebAuthn flow validation, passkey list tampering, key derivation security	High

What Makes This Audit Interesting

This is not a Solidity smart contract audit. The Options Sidechain is implemented in C++ at the rippled protocol layer — the same codebase that has secured the XRP Ledger since 2012. Auditing protocol-level financial logic in a production blockchain codebase requires a rare combination of skills:

- Deep C++ and consensus protocol expertise
- Understanding of XRPL's ledger object model, transaction processing pipeline, and invariant checking system
- Financial derivatives knowledge — options pricing, margin mechanics, liquidation cascades
- Cryptographic review capability for P256 / WebAuthn / Merkle proof verification
- Cross-chain bridge security — specifically XPop proof validation and multisig vault trust models

Funding Path

The audit will be funded through the XRPL Grants program, which supports security work that strengthens the broader XRPL ecosystem. The grant application is being prepared in parallel with outreach to audit firms. Interested firms should reach out early to discuss scope, timeline, and fit.

Firms with relevant experience in protocol-level blockchain security are encouraged to reach out: dangell@transia.co

Community Review

Three parallel review tracks are open now. We want rigorous, critical feedback before any code ships to testnet.

Track 1: XLS Specification Review

The three XLS draft specs define the complete design of the system at the protocol level. We are looking for reviewers who can identify design-level issues — ambiguities, edge cases, or security assumptions that don't hold — before they become bugs in the implementation.

Relevant questions for spec reviewers:

- Are the XPop validation steps in the Import flow complete and ordered correctly? Can any step be bypassed or reordered?
- Does the 80% quorum threshold adequately protect against validator set manipulation given realistic UNL compositions?
- Are the sealed option matching semantics correct for all exercise, close, and expiry combinations?
- Is the cross-margin liquidation condition sufficient to prevent cascading insolvency?
- Does the passkey authentication flow correctly integrate with XRPL's existing multi-sig and regular key systems?

Specs: github.com/Transia-RnD/rippled/tree/options-sidechain/specs

Track 2: C++ Code Review

The implementation lives in the options-sidechain branch of the rippled repository. This is production C++ code running the full XRPL transaction processing pipeline. Code reviewers should focus on:

- Correctness of XPop Merkle proof verification and ledger hash computation
- Margin math implementation — integer overflow, rounding errors, and precision in fixed-point arithmetic
- Invariant check coverage in MarginInvariant.h — are all edge cases caught?
- Object lifecycle management for ExportRecord, MarginPosition, and SealedOptions
- Error handling paths — particularly in the Import preflight/preclaim/doApply split
- Memory safety and any potential for denial-of-service via malformed XPop blobs

Branch: github.com/Transia-RnD/rippled/tree/options-sidechain

Track 3: Economic & Game Theory Review

Financial systems fail in ways that code reviews don't catch. We want economists and DeFi researchers to stress-test the incentive structures and identify attack vectors at the economic layer:

- Can the insurance vault be drained through coordinated liquidation timing?
- Are funding rates sufficient to prevent long/short imbalances from becoming structural?
- Can a large naked option seller create systemic risk if liquidated simultaneously with volatile price movement?
- Does the 1:1 lock-and-mint model create any reflexivity risks if XRP price moves sharply during a bridge transit?
- Is the liquidation bonus sized correctly — large enough to incentivize liquidators but small enough to avoid value extraction from position owners?
- Can the option book be manipulated via wash trading to influence mark price for liquidation purposes?

How to Contribute

Open a GitHub issue on the options-sidechain branch with your findings, or reach out directly:

dangell@transia.co • github.com/Transia-RnD/rippled

All meaningful contributors will be acknowledged in the final XLS specification and in the genesis documentation of the sidechain.

Get Involved

Three ways to help build the first native derivatives layer on the XRP Ledger:



Run a Validator

Established XRPL UNL validators. Extend your infrastructure to the options sidechain and join the founding validator set.



Audit the Code

Professional security firms with protocol-level blockchain and financial systems experience. Grant-funded via XRPL Grants.



Review the Specs

XRPL developers, economists, and DeFi researchers. Three open tracks: XLS design, C++ code, and economic security.

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This is early. That's the point.

The validators, auditors, and reviewers who engage now will shape what on-chain derivatives looks like for XRPL.